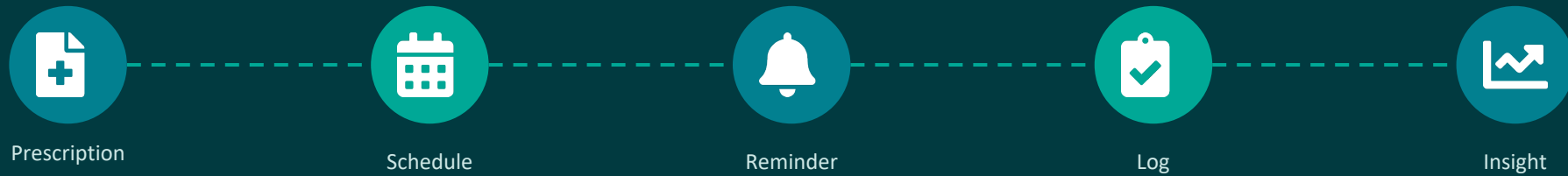


MediSync

Smart Medication Management & Adherence



Group Members:

- Raghad Almohammadi
- Jana Al-khalifa
- Lujin Alsalmi
- Nada Aljuhani
- Raghad Almalaq
- Rana Alahmadi

THE PROBLEM

Managing Medications Gets Complicated, Fast

As treatment plans grow, patients must track more than just "which pill" — they must track how, when, and under what conditions.



Multiple medications



Different times of day



Weekly / biweekly doses



Potential drug interactions



Different doses



Different frequencies



Before / after food rules



Overlapping schedules

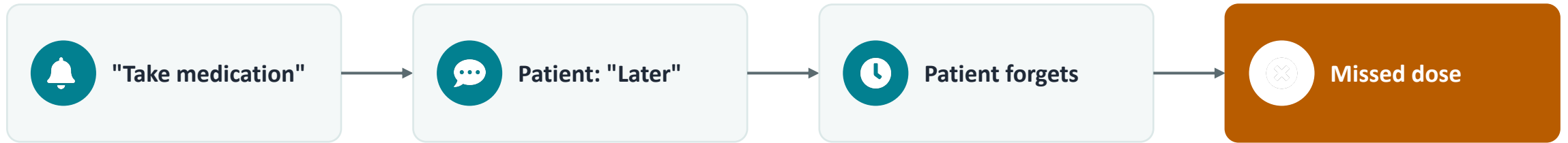
When it gets this complex, patients may:

Forget

Postpone

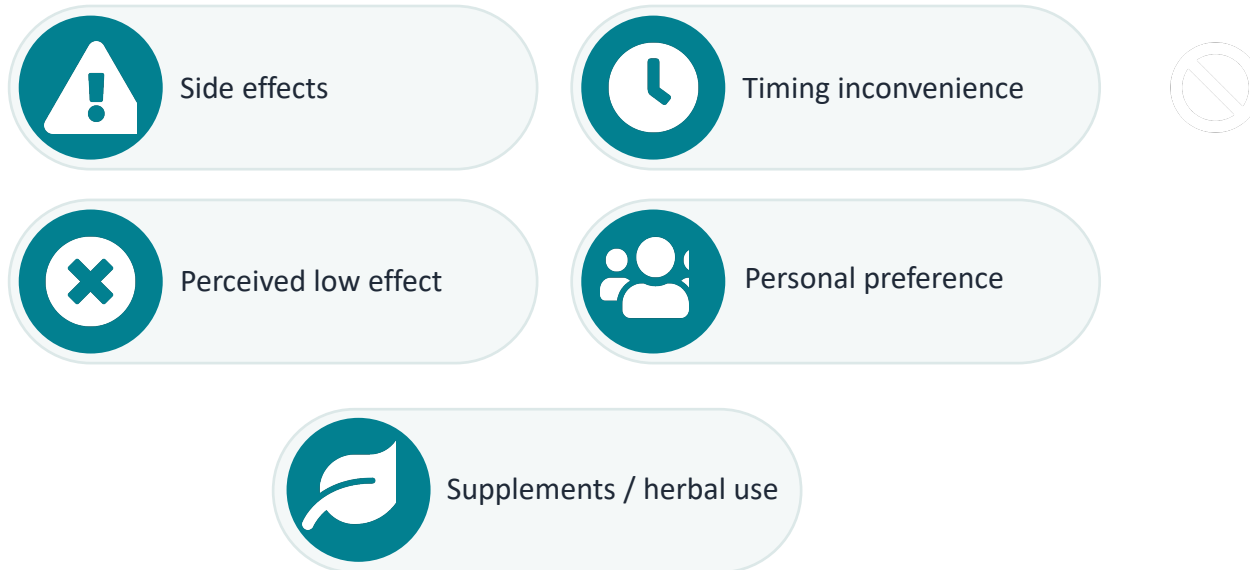
Stop

Why Reminders Alone Are Not Enough



*A reminder guarantees a notification — **not adherence.***

Patients often stop or skip medications for reasons unrelated to forgetting:



Key idea:

the system should understand medication-taking behavior — not simply send reminders.

Grounded in Established Healthcare Research



Regimen Complexity

More complex medication schedules — multiple drugs, doses and timings — are consistently linked to greater difficulty in maintaining adherence.



Medication Non-Adherence

A 2024 retrospective review published in *Frontiers in Pharmacology* confirms that global chronic medication non-adherence remains unchanged at 30% to 50%.



Electronic Reminders

Reminder-based tools can support adherence, but reviews of digital interventions show reminders alone do not address the underlying causes of non-adherence.



Medication Safety

Combining prescription, over-the-counter and herbal products without oversight increases the risk of interactions and adverse effects.

Sources: *Frontiers in Pharmacology* (2024/2025), "Medication nonadherence" (2003) • Peer-reviewed systematic reviews (PubMed) • Saudi Food & Drug Authority • Saudi Ministry of Health

Who MediSync Is Built For

PRIMARY USERS



Patients on multiple medications

Need help organizing complex, overlapping regimens.



Patients with chronic conditions

Rely on consistent, long-term adherence for treatment to work.



Healthcare providers

Gain adherence insight to inform follow-up and clinical decisions.

What MediSync Aims to Deliver

The project is designed around one core objective — turning medication-taking into a supported, understood process — with benefits for every stakeholder.

FOR PATIENTS



One clear, organized medication schedule



Reminders that adapt to their routine



A simple way to log taken, missed, or stopped doses

FOR HEALTHCARE PROVIDERS



Adherence patterns, not just isolated events



Insight into why doses are missed or stopped



Better-informed follow-up conversations

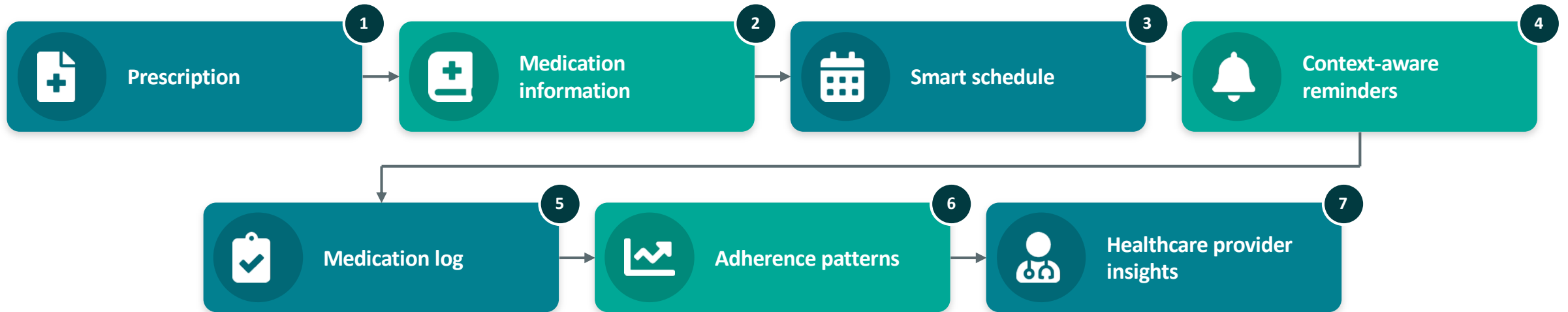
Relevance to the Saudi Healthcare Ecosystem

Saudi Arabia already offers strong digital healthcare infrastructure:

 Sehaty — health services & records

 Wasfaty — e-prescription dispensing

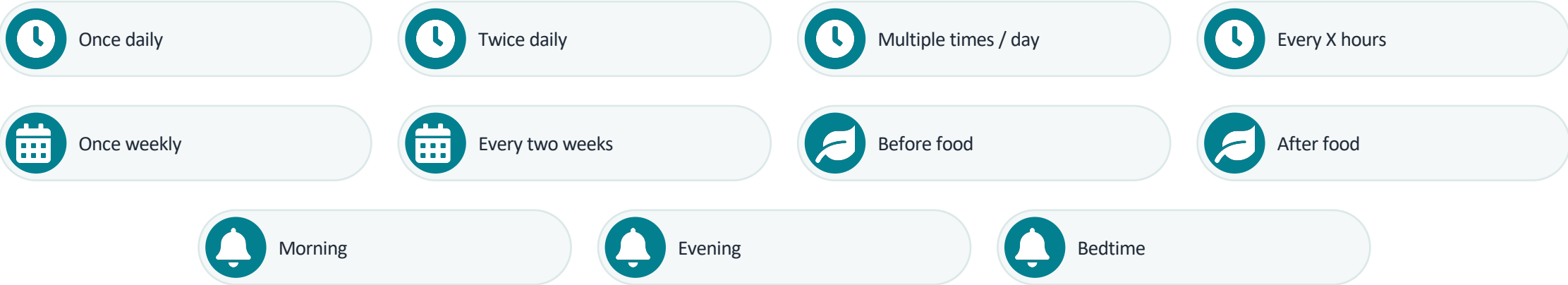
❖ transforms a prescription into an organized, trackable treatment plan.



Each stage of the flow maps directly to a set of software features and data entities developed across the semester.

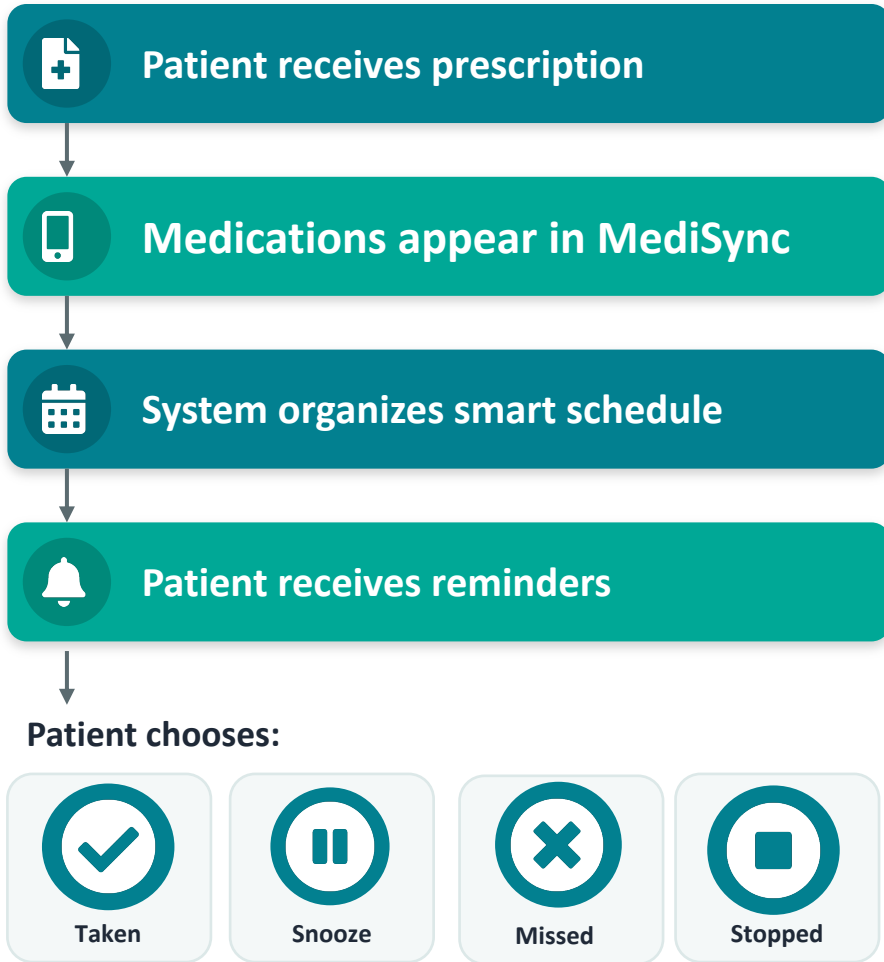
Organizing Complex Medication Instructions

MediSync converts varied dosing instructions into a single, organized schedule per patient.



The system organizes timing and flags potential conflicts using trusted medication information — it does not independently make clinical decisions.

How the Patient Uses MediSync



If a dose is missed or stopped:

- Patient provides a short reason**
- Becomes part of the adherence history**

This turns everyday medication-taking into structured adherence data the patient and provider can both use.

Reason-Based Logging

Traditional Reminder App



Missed dose

...and that is all the system knows.

MediSync



Missed dose



Why?

Forgot

Side effects

Timing issue

Unavailable

Cost

Using another
product

Other

Healthcare providers see not just that a dose was missed, but the pattern of why.

What a Healthcare Provider Could See



Patient: Example Patient

Most reported reason: Side effects

Medication A

85%

Adherence



4

Missed

6

Snoozed

0

Stopped

Medication B

62%

Adherence



7

Missed

0

Snoozed

3

Stopped



MediSync does not prescribe or automatically change medications — it gives providers information to support their own clinical review.



“MediSync transforms medication management from a simple reminder system into a connected adherence-support system — helping patients follow their treatment, and helping healthcare providers understand why adherence fails.”

Questions?